

- 21 A negatively-charged sphere P is balanced halfway between two horizontal plates when a potential difference V is applied between the plates.

Which statement is correct?

- A Increasing V increases both the electric and the gravitational potential energy of the sphere.
- B Increasing V decreases the electric potential energy and increases the gravitational potential energy of the sphere.
- C Decreasing V decreases both the electric and the gravitational potential energy of the sphere.
- D Decreasing V increases both the electric and gravitational potential energy of the sphere.

Ans: B

When the potential difference is increased, the sphere will experience a greater upward electric force. Therefore, the electric force is bigger than the weight of the sphere. Net force will be upwards and hence sphere will accelerate upwards, and thus experiencing a decreasing EPE and increasing GPE.

(Answer will be the same even when the sphere is positive charged)

22. A 9.0 A current passes through a cylindrical copper wire with a diameter of 0.0 mm. The density